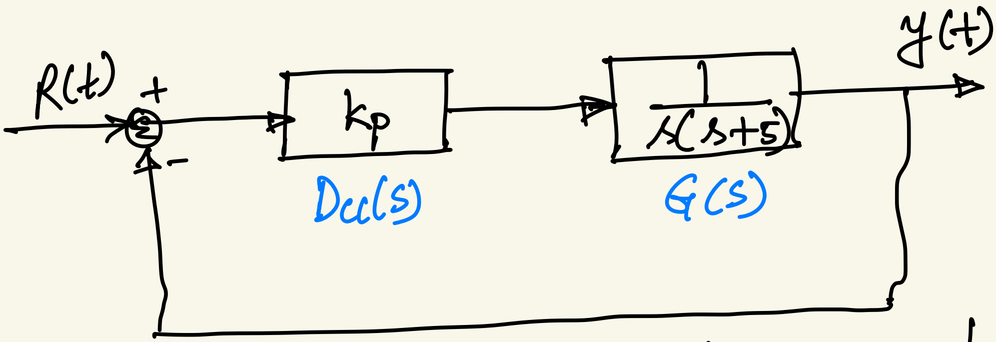
Example:

Design the value of gain k_p for the following system $G(s)$ so that the system will respond with a 10% overshoot.

$$G(s) = \frac{1}{s(s+5)}$$



$D_{cl}(s) = k_p$, proportional controller

$$G(s) = \frac{1}{s(s+5)}$$

$$T_{cc}(s) = \frac{D_{cc}(s)\, G(s)}{1 + D_{cc}(s)\, G(s)}$$

$$= \frac{k_p \cdot \frac{1}{s(s+5)}}{1 + k_p \cdot \frac{1}{s(s+5)}}$$

$$= \frac{\frac{k_p}{(s^2+5s)}}{\frac{s^2+5s+k_p}{(s^2+5s)}}$$

$$T_{cc}(s) = \frac{k_p}{s^2+5s+k_p}$$

We compare with the second order transfer function

$$T(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$2\xi\omega_n = 5 \quad \text{--- (i)}$$

$$\omega_n^2 = K_p \quad \text{--- (ii)}$$

$$\Rightarrow \omega_n = \sqrt{K_p}$$

For 10% overshoot

$$e^{-\xi\pi/\sqrt{1-\xi^2}} = 0.1$$

$$\Rightarrow \xi = 0.59$$

$$2(0.59)\sqrt{K_p} = 5$$

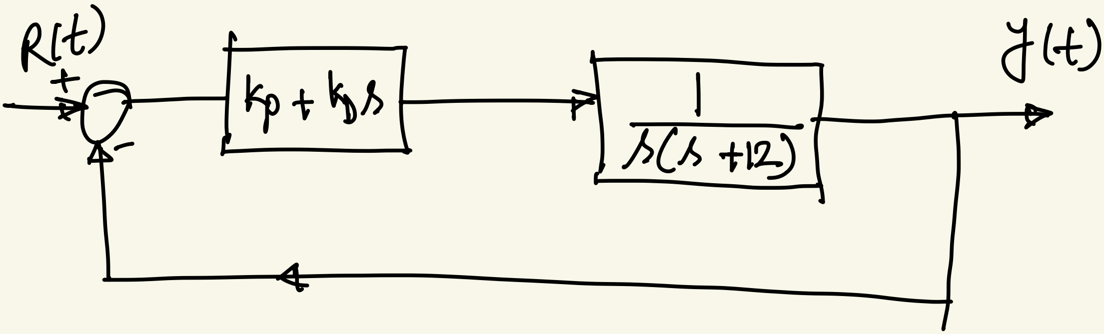
$$\Rightarrow K_p = 17.89$$

Example:

PD controller design:

Given plant: $G(s) = \dfrac{1}{s(s+12)}$

For a ramp input, design a PD controller so that the steady state error is less than 1% and damping ratio $\zeta = 0.707$



$$D_{cc}(s) = K_p + K_D s$$

$$G(s) = \frac{1}{s(s+12)}$$

$$T_{cc}(s) = \frac{D_{cc}(s)\, G(s)}{1 + D_{cc}(s)\, G(s)}$$

$$= \frac{\dfrac{(k_p + k_D s)}{s(s+12)}}{1 + \dfrac{(k_p + k_D s)}{s(s+12)}}$$

$$= \frac{(k_p + k_D s)}{s^2 + 12s + k_D s + k_p}$$

$$T_{cc}(s) = \frac{(k_p + k_D s)}{s^2 + (12 + k_D)s + k_p}$$

steady state error, $e_{ss} = \dfrac{1}{k_v}$

where, $k_v = \lim_{s \to 0} s\, G_{op}(s)$

Open loop transfer function

$$G_{op}(s) = \frac{k_p + k_d s}{s(s+12)}$$

$$k_v = \lim_{s \to 0} \cancel{s} \cdot \frac{k_p + k_d s}{\cancel{s}(s+12)}$$

$$= \lim_{s \to 0} \frac{k_p + k_d \cdot 0}{0 + 12}$$

$$k_v = \frac{k_p}{12}$$

$$e_{ss} = \frac{1}{k_v} = \frac{12}{k_p} = 0.01$$

$$\Rightarrow k_p = 1200$$

By comparing with the 2nd

order transfer function.

$$k_p = 1200$$

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = s^2 + (12 + k_D)s + 1200$$

$$\Rightarrow \omega_n^2 = 1200$$

$$\Rightarrow \omega_n = \sqrt{1200} = 34.64$$

$$12 + k_d = 2\zeta\omega_n = 2(0.707)(34.64)$$

$$\Rightarrow k_d = 37$$

$$\boxed{D_{c1}(s) = 1200 + 37s}$$